Machine Learning Models for County-Level Analysis of Voter Turnout & Partisan Lean in the 2024 U.S. Presidential Election

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Abstract

Partial election results can be used to call the winner of an election before all votes have been counted. Linear regression methods and an artificial neural network were trained on election and demographic data to predict the Democratic vote share and total turnout in various counties for the 2024 election. To simulate a scenario where partial returns are used to predict final election results, the test set was manually selected to be counties in states in the western United States that report their results later in the night. The best-performing model correctly predicted the winner of each state in the test set. Additionally, random forest models achieved high accuracy in predicting whether a county in the test set would vote for a Democrat or have above-average voter turnout.

Keywords: 2024 Election Prediction, Voter Turnout Prediction, Artificial Neural Networks, Best Subset Selection, Random Forest, Logistic Regression

1 Introduction

Following the conclusion of the 2024 election cycle, the use of statistical methods to forecast and understand election results faces a crisis of confidence. The bulk of the criticism has been directed at survey research, which has consistently underestimated Donald Trump's strength in the past three presidential elections. The media and the general public have lost much faith in the accuracy of polling to give an accurate prediction of election results, epitomized by the failure of J. Ann Selzer's "gold standard" poll of Iowa in the final days leading up to the 2024 election.[1] While her survey predicted a three-point lead for Kamala Harris among Republican-leaning Iowans, shocking the political sphere and leading some Democrats to expect a decisive victory on election night, the Hawkeye State ultimately delivered a 13-point margin for Trump. Trump's strong performance in Iowa, despite Selzer's prediction, led him to file a lawsuit alleging "election interference" in an attempt to defraud voters into believing Harris' victory in the state was inevitable.

Frivolous or not, the lawsuit reflects a growing trend among political figures in stoking voters' existing mistrust of data and statistics in the political sphere, thereby creating confusion and uncertainty about the future. Election night in 2020 served as a case study for the threat posed by the combination of uncertainty about election results and a political leader lacking a commitment to democratic institutions. In the unique circumstances of the COVID-19 pandemic, the electorate utilized mail balloting and early voting to an unprecedented degree, resulting in delays in vote reporting by election officials that allowed Trump to construct a narrative about the election being rigged in favor of Joe Biden.

At the same time, media organizations use pre-election polling and partial election returns to make projections on election night, allowing viewers to understand who will win an election without having to

wait for every vote to be counted. Organizations like the Associated Press and television networks use statistical modeling techniques to understand how the results from early-reporting localities might change their pre-election priors about who is favored to win an election and determine which candidate still has a path to victory based on the votes that remain to be counted. In close elections, this is often easier said than done. The television networks infamously called Florida for Al Gore early in the night in 2000 before retracting the decision.[2] In 2020, Fox News' early call of Arizona for Biden caused a revolt by network executives, and at times, the published forecasts of how states would vote were clearly miscalibrated.[3] The chaos at decision desks, as well as the delay in making calls due to the unknown impact of early and mail ballots, only compounded the perceived uncertainty of the 2020 election results, providing an opportunity for Trump and his allies to sow doubt in the validity of the entire electoral process.

Given the importance of accurate and timely projections on election night, it is pertinent to consider the various methodologies available to stakeholders who may be interested in knowing the final election results. News outlets have an evident and public-facing need for accurate election results, both to provide accurate information to viewers and to enhance their reputation for making fast projections. Campaigns (operating in good faith) need to understand their chance of winning to hone their messaging, and in close races, allocate resources in the event of a recount or litigation. A new entrant into the space are betting companies and their patrons, who rely on accurate signals of how election night will play out.

Designers of election night models have an interest in maintaining secrecy behind the inner workings of their models, so the information available to the general public is vague. For instance, the New York Times describes its Needle, which provides readers with live projections of win probabilities and margins, as follows:

> As results begin coming in, the Needle compares what is being reported with pre-election expectations, county by county and precinct by precinct. It then estimates who will win the remaining vote based on patterns it has seen in the results so far. [4]

The Times further specifies that it incorporates demographic data to infer that if a candidate is, for example, performing well in highly educated counties, the Needle will adjust its expectations in other highly educated counties in favor of said candidate. However, there is no further information about the mathematics behind how the Needle combines its information streams to produce its estimates. As such, this analysis will develop a high-level, open model for training a model during an election.

This analysis utilizes publicly available data on each county in the United States to predict the outcome of the 2024 election in that county. Different data sets containing information for each county on 2020 election results, 2024 election results, education, poverty, unemployment, income, population, and race were combined to create a master dataset. The idea is that, using demographic information and 2020 election results, an accurate model could be created that predicts 2024 election outcomes. To simulate a real election night, the model will be trained on states that report their results early and then tested on the remaining states.

Some of the research questions that this project will address include:

- How accurately can the percentage of Democratic votes in each county for 2024 be predicted?
- How accurately can the winner of each county for 2024 be predicted?
- How accurately can the total number of votes in each county for 2024 be predicted?
- What are the most important variables in predicting the percentage of Democratic votes in each county for 2024?

2 Data Collection

Demographic data at the county level is readily available from several sources. The United States Department of Agriculture (USDA) maintains a database of county-level demographic data from different Census data sources.[5] The USDA dataset contains four broad categories of data. First, the data include information about income and poverty levels for each county and county-level equivalent in the United States. Alaska, Connecticut, and the District of Columbia were excluded from the dataset since the jurisdictions for which election data and demographic data were available differed. In particular, two selected variables were coded as `poverty` (the point estimate for the poverty rate in a county during the period from 2017 to 2023) and `mhi` (the point estimate for the median household income in a county during the period from 2017 to 2023). Second, the USDA dataset contains population data, which were coded as pop (the population of a county in the

2020 Census), `rural_urban` (a variable where 1 indicates the most urban and 9 indicates the most rural), `pop_chg` (the percent change of a county's population in 2023), and `net_migration` (the net domestic migration as a percentage of a county's population in 2023). There is also a dataset containing information on unemployment, which was used to calculate the `unemployed` variable (the point estimate of the unemployment rate in a county in 2023). Finally, the USDA dataset contained a subsection on education rates, which was used to construct `pct_no_hs` (the proportion of a county's adult population without a high school diploma), `pct_hs` (the proportion of a county's adult population whose highest education is a high school diploma), and pct_bach (the proportion of a county's adult population who have a Bacholor's Degree or a higher level of education).

Additionally, census data were collected about the racial demographics of each county. In particular, the American Community Survey's 2023 five-year estimates were used to construct the variables `hisp` (proportion of a county identifying as Hispanic or Latino), `white` (the proportion of a county identifying as non-Hispanic White), `black` (the proportion of a county identifying as non-Hispanic Black or African American), `namerican` (the proportion of a county identifying as non-Hispanic American Indian or Native Alaskan), `asian` (the proportion of a county identifying as non-Hispanic Asian), and mixed (the proportion of a county identifying as two or more races.[6]

Finally, election data were pulled from a repository maintained by Tony McGovern, who has scraped election results from 2008 to 2024 from different news sites such as Politico, Fox News, and The New York Times. [7] Although these sites are reputable, the results may differ slightly from certified election results available directly from the election authorities of each state. A spot check comparing the repository's election results with certified results did not show significant or systematic discrepancies. This dataset was used to construct the variables `total_votes_20` (the total number of votes cast in a county in the 2020 presidential election), `per_dem_20` (the share of the vote received by Democrat Joe Biden in the 2020 presidential election in each county), `total_votes_24` (the total number of votes cast in a county in the 2024 presidential election), and `per_dem_24` (the share of the vote received by Democrat Kamala Harris in the 2020 presidential election in each county).

These three datasets were combined according to their Census FIPS code, a unique identifier for each county in the United States. Each row of the dataset represents a county, and each of the previously described variables is a column alongside the additional identifiers of state and county. Finally, the data were manually split into training and test sets based on the timing of each state's election results reporting. States can only begin to report their election results once voting locations have closed. For instance, the first results are always reported from Indiana and Kentucky, where the majority of voting sites close at 6 PM Eastern. To simulate an election night where the election results from early states are used to predict outcomes in states with few or no votes counted, states with early poll closing times were manually selected as the training set, and states with late poll closing times were manually selected as the test set. Some key swing states, such as Wisconsin and Michigan, were also selected for the test set to assess how the predictive models would perform in states where the election was likely to be close. The only other traditional swing states with the latest poll closing times are Arizona and Nevada. Ultimately, the training set comprises 2,217 counties, while the test set consists of 886 counties.

3 Methods

3.1 Best Subset Selection

After data pre-processing, the next step was to identify the most important features for further modeling. To determine the key predictors of the 2024 Democratic vote share, the dataset was first cleaned by removing the `total_votes_24` column, as this information is not reasonably known beforehand. Then, a best subset selection approach was applied using the regsubsets() function from the leaps package, which tested combinations of variables to find the best-fitting linear model. Although forward selection, backward selection, and best subset selection were all considered, best subset selection—a method that evaluates all possible predictor combinations to identify the most effective model—was chosen due to the manageable number of features, which made the approach computationally feasible.

Model performance was evaluated using different metrics, and plots were created to visualize these values across models with different numbers of predictors.

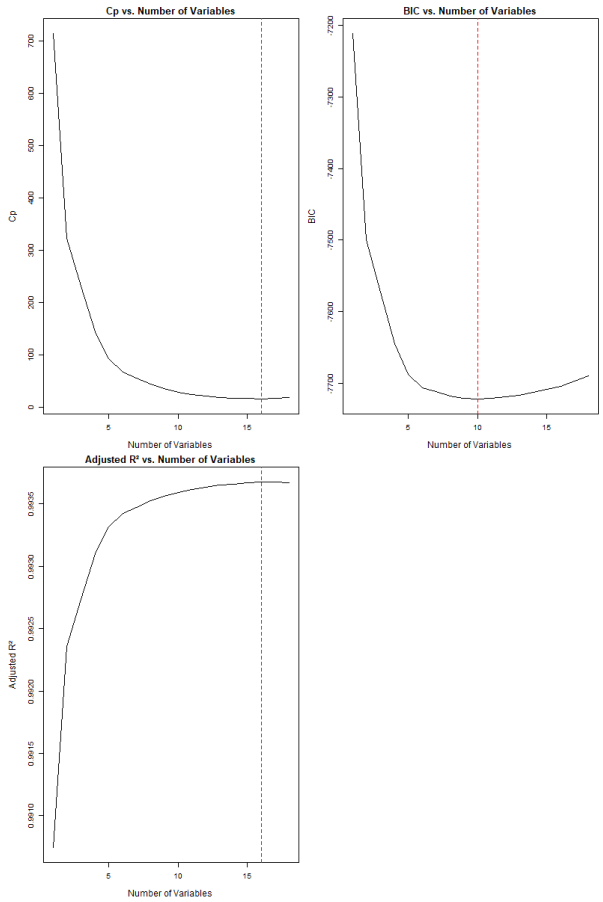

Figure 1: Model selection metrics vs. number of variables

As can be seen in Figure 1, the best Mallows' Cp corresponds to the 10-feature model. Mallows' Cp measures the trade-off between model complexity and fit, with lower values indicating models that are both accurate and not overly complex. Also seen in Figure 1, the best adjusted R^2 corresponds to the 10-feature model. Adjusted R^2 measures the proportion of variance explained by the model, adjusted for the number of predictors. Therefore, higher values of R^2 correspond to a better-fitting model. Finally, also seen in Figure 1, the best BIC (Bayesian Information Criterion) corresponds to the 10-feature model. BIC measures model fit while applying a more substantial penalty for complexity than other metrics. Lower BIC values indicate a better balance between fit and model simplicity, helping to prevent overfitting. Because of this, BIC was chosen as the primary metric for modeling, selecting the optimal subset for the 10-feature model to use in future modeling. After identifying the best subset of variables, both the training and test datasets were updated to include only these selected features. The cleaned and reduced datasets were saved for future modeling.

The most important variables that were kept are: percentage of democrat votes 2020, percent in poverty, percent no high school degree, percent with highest education being high school, percent with highest degree being bachelors or higher, population, percent unemployed, percent Black, percent Hispanic, percent Asian, and percent reporting two or more races.

Variables excluded during gesture selection due to their relatively lower predictive value include: total votes in 2020, rural-urban classification, median household income, percentage of white residents, percentage of Native American residents, percentage change in population, and percentage change in migration.

3.2 Linear Regression

Four versions of a linear regression model were used to predict Kamala Harris's vote share and the total turnout in each county in states that report election results later in the night. First, a linear regression model was fit to the entire training set. This method reduces the variance of the coefficient estimators, but it may result in overfitting and an inflated bias when the model is used to predict turnout or Harris's vote share for the test set. Second, a linear regression model was fitted using 5-fold cross-validation, where the model with the lowest validation error was selected to predict Harris's vote share and turnout in the test set. Using cross-validation reduces the risk of overfitting since the effect of outliers on coefficient estimates is mitigated. Third, linear regression was fit to the entire training set using the variables selected via best subset selection. It is essential to note that best subset selection was applied only to the Harris vote share variable, and therefore, the best subset linear regression performed poorly in predicting turnout. Finally, a LASSO regression model was fitted using the best lambda value found through cross-validation. The LASSO regression is more flexible in zeroing out variables that have no significant effect on the predictor variable. However, the magnitude of the coefficients may be artificially reduced, possibly resulting in underfitting. The benefit of regression models lies in the straightforward interpretation of variables, as each coefficient represents the effect of a one-unit change in a predictor variable on the corresponding change in the dependent variable.

Models were evaluated using Mean Absolute Error (MAE), which measures the mean of the absolute differences between the predicted values and the actual values of the test set, and Root Mean Squared Error, which measures the square root of the mean of

the squared differences between the predicted values and the actual values of the test set.

3.3 Artificial Neural Network (ANN)

The next model used is an artificial neural network (ANN) to perform three prediction tasks: predicting the percentage of Democratic votes (`per_dem_24`) as a regression problem, classifying whether the Democratic candidate won in a county, and predicting the total number of votes cast (`total_votes_24`). The best subset was used that was identified in the earlier selection process on all ANN models. For the classification task, a binary variable `dem_winner` was created based on whether the Democratic vote share exceeded 50%. This replaced `per_dem_24` as the model target.

Input features were first standardized, using the `StandardScaler` package to ensure that they had zero mean and unit variance. This step is critical as it helps accelerate convergence and improves stability during optimization. The training data was then divided into a training set and a validation set using an 80/20 split. The validation set is a portion of the data not seen by the model during training, used to monitor how well the model generalizes to unseen data. Early stopping, a technique that stops training when performance on the validation set stops improving, helping to avoid overfitting. If early stopping was not implemented, the model might perform almost perfectly on the training data but perform significantly worse on the test data, making the model much less useful through overfitting.

The actual architecture of the ANN included two hidden layers, each having 64 neurons applied with a ReLU (Rectified Linear Unit) activation function. Activation functions are crucial to any neural network as they introduce non-linearity into the network that enables the model to learn patterns beyond simple linear trends.

Using manual hyperparameter tuning, model performance was improved using dropout regularization, batch normalization, and L2 regularization. Dropout regularization is a regularization method that randomly disables some neurons during training to prevent over-reliance on specific weights. Batch normalization is a normalization method in which normalization is applied to each of the inputs to each layer during training, helping to create a stable model network. L2 regularization is a regularization technique used to penalize large weights in the model, further reducing the risk of overfitting. The strengths of these techniques were manually tuned to find the combination that produced the best model performance.

The model was compiled using the Adam optimizer, an adaptive learning algorithm that adjusts learning rates during training to achieve faster and more stable convergence.

For the classification task, we used binary cross-entropy (BCE) loss, which is the common loss function used for binary classification tasks. BCE measures the difference between predicted probabilities and actual binary labels. To assess performance, we evaluated the model on the test set using standard classification metrics:

- Accuracy, which measures the overall proportion of correct predictions
- Precision, which measures the proportion of true positive predictions among all positive predictions
- Recall, which measures the proportion of true positives identified out of all actual positives
- F1-score, which balances precision and recall

A confusion matrix was also made, which is a plot that shows the counts of true positives, true negatives, false positives, and false negatives to give a full picture of the prediction performance.

For our regression tasks, we used the mean absolute error (MAE) as the loss function. MAE is a common metric for regression problems as it measures the average magnitude of errors between predicted and actual values, without considering their direction. To assess performance, we evaluated the model on the test set using MAE as well.

3.4 Binary Classification of Partisan Lean & Voter Turnout

Two binary classification tasks were conducted: 1) Democratic-leaning vs. non-Democratic-leaning counties based on 2024 vote share (`per_dem_24`), and 2) high vs. low county-level voter turnout in the 2024 presidential election (`total_votes_24`). For each task, the median value of the target variable in the training set was used to assign binary labels, where 1 indicated values above the median and 0 indicated values below. The median was chosen over the mean to reduce sensitivity to outliers and skewed distributions.

All features were standardized using z-score normalization (StandardScaler). For the Democratic-leaning classification, feature selection was conducted in R using the best subset selection method based on the Bayesian Information Criterion (BIC). For the turnout classification, forward stepwise feature selection (SequentialFeatureSelector) was applied in

Python using logistic regression as the base model. The following models were fit for both classification tasks:

- Logistic Regression
- Random Forest
  (`n_estimators` $\in [10, 50, 100, 200, 300]$)
- Support Vector Machine
  ($C \in [0.001, 0.01, 0.1, 1, 10, 100]$)
- K-Nearest Neighbors ($k \in [1, 20]$)

These models were selected to represent a range of linear and nonlinear classifiers with varying levels of complexity and interpretability. Model performance was evaluated using accuracy, precision, recall, and F1 score. Validation curves were generated via 5-fold cross-validation. All modeling was implemented in Python using `scikit-learn`, `pandas`, `numpy`, and `plotly`.

4 Results

Table 1: Test Errors for Linear Regression Models

Model	RMSE (H%)	MAE (H%)	RMSE (Turnout)	MAE (Turnout)
Linear Regression	1.450	1.127	12095.91	3447.34
Linear Regression (CV)	1.428	1.102	11820.92	3367.71
Best Subset	1.428	1.111	25198.50	11270.87
LASSO	1.413	1.100	12212.26	3598.61

MAE is more robust to outliers than the RMSE and will thus be the metric used to select the best model. LASSO had the lowest MAE for predicting Kamala Harris's vote share in a given county from the test set. A MAE of 1.10% suggests that, on average, the LASSO model was able to predict Harris's vote share to within 1.1% of her actual vote share in a county. All the models performed similarly, with the highest MAE of 1.13% only negligibly different from the lowest MAE.

Linear Regression with Cross-Validation performed best at predicting turnout. The MAE of 3,367 means this model can predict the total turnout in a county to within 3,367 votes, on average. Every model but the Best Subsets had an MAE below 3,600 votes. The Best Subsets regression had a high MAE (11,270 votes) since its variables were selected to predict Democratic vote share rather than turnout.

Table 2: Coefficients of LASSO Regression of Harris Vote Share

Variable	Coefficient
`(Intercept)`	0.0143
`total_votes_20`	0.0000
`per_dem_20`	0.9552
`rural_urban`	-0.0002
`poverty`	-0.0002
`mhi`	0.0000
`pct_no_hs`	-0.0005
`pct_hs`	-0.0002
`pct_bach`	0.0003
`pop`	0.0000
`unemployed`	-0.0010
`hisp`	-0.0003
`white`	0.0000
`black`	0.0001
`namerican`	0.0000
`asian`	-0.0006
`mixed`	0.0004
`pct_change`	0.0000
`pct_migration`	-0.0240

LASSO coefficients are biased towards zero, and thus the coefficient values do not have the exact interpretation that can be obtained via ordinary least squares (OLS) regression. There is no direct way to ascribe statistical significance to a result; instead, the emphasis falls on the predictive influence and power of a variable. It is clear that the most significant predictor of Harris's vote share in 2024 is Biden's vote share in 2020 in the same county. A one-standard-deviation increase in Biden's vote share results in an increase of Harris's vote share by 95.5%, on average, holding other variables constant. Other variables with large coefficients include net domestic migration as a proportion of a county's population, which is negatively correlated with Harris's vote share, as is the proportion of the county's population that is unemployed. Political polarization likely explains the strong correlation between the vote shares of Biden and Harris. A higher unemployment rate in a county could cause voters to sour on the incumbent administration, while the significance of net migration could be related to the disproportionate population growth in conservative states like Texas and Florida.[**?**]

Table 3: Coefficients of Cross-Validated Linear Regression of Total Turnout

Variable	Coef.	SE	t-stat	p-val
(Intercept)	61570.881	41997.031	1.466	0.143
total_votes_20	1.133	0.024	47.696	0.000
per_dem_20	-1913.432	1773.434	-1.079	0.281
rural_urban	-135.130	65.098	-2.076	0.039
poverty	-43.796	40.970	-1.069	0.286
mhi	0.023	0.021	1.080	0.281
pct_no_hs	-32.228	40.316	-0.799	0.425
pct_hs	-5.163	34.735	-0.149	0.882
pct_bach	-34.683	40.918	-0.848	0.397
pop	-0.086	0.012	-7.437	0.000
unemployed	164.348	172.040	0.955	0.340
hisp	-584.434	419.049	-1.395	0.164
white	-597.527	420.069	-1.422	0.156
black	-579.646	420.105	-1.380	0.168
namerican	-632.446	425.144	-1.488	0.138
asian	-336.739	441.068	-0.763	0.446
mixed	-488.254	439.573	-1.111	0.267
pct_change	156953.635	43855.041	3.579	0.000
pct_migration	-81292.921	43930.246	-1.851	0.065

There is a near-perfect linear correlation between 2020 turnout and 2024 turnout, with a one-vote increase in the number of votes cast in 2020 corresponding to a 1.13-vote increase in the predicted votes cast in 2024, assuming all other factors remain equal. The other significant predictors are the total population and the percentage change in population from 2022 to 2023. As expected, faster-growing counties also tended to have more votes cast after accounting for the other variables. The total population of a county was actually negatively correlated with turnout. Since turnout in 2020 has been accounted for in the regression, the negative correlation suggests that larger counties experienced a drop in turnout from 2020 to 2024, all else being equal.

The best-performing models (LASSO for Harris's vote share, cross-validated linear regression for turnout) can be combined to create state-level predictions of Harris's vote share. Multiplying the predicted Democratic share of the vote in a county by the predicted number of votes in a county results in an estimate of the number of votes Harris receives in a county. These estimates are then summed and divided by the predicted total number of votes in a state, resulting in a prediction of Harris's vote share in each state of the test set.

Table 4: Predictions of Cross-Validated Linear Regression of Total Turnout

State	Pred. Harris Votes	Pred. Turnout	Pred. Harris %	Harris Votes	Turnout	Harris %
Arizona	1,466,835.61	3,241,816.8	45.25	1,582,860	3,389,319	46.70
California	9,481,428.37	16,650,786.1	56.94	9,276,179	15,862,536	58.48
Colorado	1,767,307.76	3,311,414.7	53.37	1,728,159	3,190,873	54.16
Hawaii	327,606.63	563,121.9	58.18	313,044	516,701	60.59
Idaho	300,516.75	969,278.9	31.00	274,972	904,812	30.39
Iowa	744,825.92	1,714,224.2	43.45	707,278	1,656,849	42.69
Michigan	2,638,645.57	5,427,076.6	48.62	2,736,533	5,662,504	48.33
Montana	259,809.88	683,320.6	38.02	231,906	602,963	38.46
Nevada	616,878.15	1,357,904.0	45.43	705,197	1,484,840	47.49
New Mexico	461,513.80	913,062.8	50.55	478,802	923,403	51.85
North Dakota	116,546.59	391,818.1	29.75	112,327	365,059	30.77
Oregon	1,267,288.29	2,339,060.7	54.18	1,228,410	2,207,592	55.64
South Dakota	155,416.73	465,031.0	33.42	146,859	428,922	34.24
Tennessee	1,096,720.47	3,114,084.5	35.22	1,056,265	3,063,942	34.47
Utah	538,596.52	1,514,876.2	35.55	562,566	1,487,951	37.81
Washington	2,210,885.44	4,025,675.6	54.92	2,245,849	3,898,835	57.60
Wisconsin	1,567,526.72	3,279,821.9	47.79	1,668,229	3,415,213	48.85
Wyoming	81,973.34	314,560.2	26.06	69,527	266,353	26.10

The combined model accurately predicts whether or not Harris will receive the most votes in every state of the test set, including swing states like Arizona, Nevada, Michigan, and Wisconsin. If the model were used on the 2024 election night, the election could have been called for Trump before the last polls had closed.

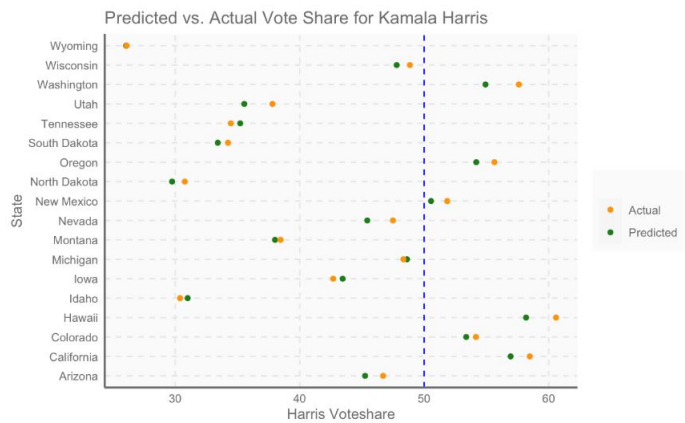

Figure 2: Predicted vs. Actual Harris Vote Share by State

4.1 Artificial Neural Network (ANN)

Table 5: Performance Metrics of ANN Classifier

Metric	Accuracy	Precision	Recall	F1 Score
Value	0.976	0.98	0.98	0.98

Table 5 shows that the ANN performed strongly, correctly classifying approximately 97.6% of counties with the actual winner of the election in that county. This result is expected given the input variables—particularly Biden's 2020 vote share—since

most counties tend to vote consistently across elections. In addition to high accuracy, the model's strong F1 score, precision, and recall indicate it is a reliable classifier.

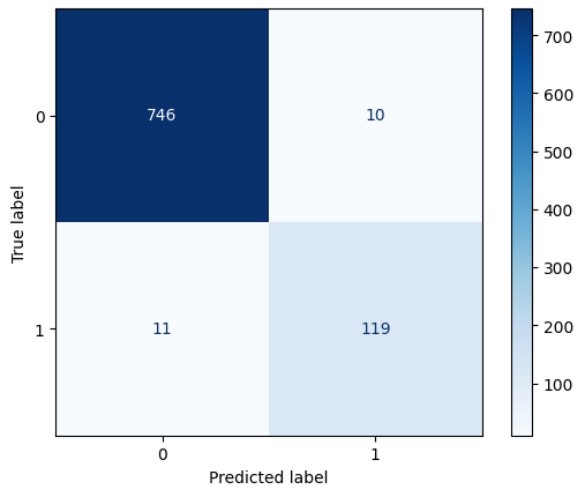

Figure 3: ANN Classification Confusion Matrix

The confusion matrix again shows that the classifier performs well overall, with high accuracy. There is a class imbalance between Democratic and Republican-leaning counties. That makes sense, as most U.S. counties are less densely populated and tend to vote Republican, while Democrat-leaning counties are fewer but more urban. This imbalance affects the model's performance metrics: F1 score, precision, and recall are all noticeably lower for Democrat-leaning counties (around 0.92) compared to Republican-leaning ones (around 0.99).

Table 6: ANN Regression Results

Test RMSE (H%)	Test MAE (H%)	Test RMSE (Turnout)	Test MAE (Turnout)
1.415	1.225	14614.728	2787.696

Table 6 shows that the ANN also performed well on the regression task, producing low error in predicting both Harris's vote share and total turnout. The test RMSE and MAE were approximately 1.42 and 1.23 percentage points for vote share, and 14,615 and 2,788 votes for turnout, respectively. These results suggest the model captured the underlying patterns effectively, likely due to strong input features and stable voting behavior across counties.

4.2 Binary Classification of Partisan Lean & Voter Turnout

4.2.1 Partisan Lean (`per_dem_24`)

All four models — Logistic Regression, Random Forest, SVM, and KNN—were evaluated on their ability to classify counties as Democratic-leaning or not, based on demographic and socioeconomic variables. Performance was assessed on the test set using accuracy, precision, recall, and F1 score (Table 7).

Table 7: Test Set Classification Metrics for Predicting whether a County Leans Democratic

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.970	0.972	0.977	0.975
Random Forest	0.974	0.970	0.987	0.979
SVM	0.951	0.979	0.940	0.959
KNN	0.867	0.928	0.844	0.884

Random Forest achieved the highest overall accuracy (0.974), outperforming Logistic Regression (0.970), SVM (0.951), and KNN (0.867). Both Random Forest and Logistic Regression exhibited strong, balanced performance, with F1 scores above 0.97, indicating effective classification of both classes. SVM showed high precision (0.979) but lower recall (0.940), suggesting the model was more conservative in assigning the Democratic-leaning label. KNN underperformed across all metrics, displaying notable overfitting with a significant gap between training and validation scores for varying values of k (Figure 8).

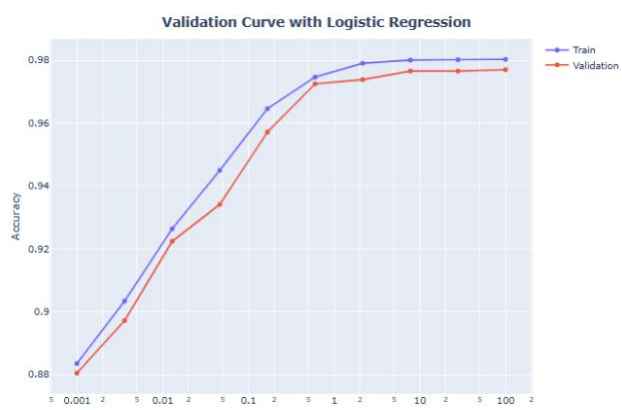

Figure 4: Validation Curve for Logistic Regression Predicting Democratic-Leaning Counties

Validation curves were necessary to highlight the key differences in model behavior and generalization. For Logistic Regression (Figure 4), accuracy improved consistently with increasing C, reaching a plateau of approximately 0.98 for both the training

and validation sets, indicating effective regularization and generalization. SVM followed a similar pattern (Figure 5) but with slightly reduced validation performance, indicating marginally less stability.

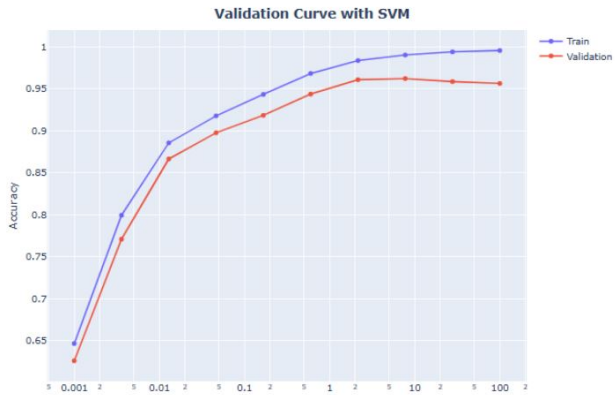

Figure 5: Validation Curve for SVM Predicting Democratic-Leaning Counties

Random Forest (Figure 6) exhibited saturated training accuracy across all n_estimators, whereas validation accuracy remained lower and flat, suggesting potential overfitting despite strong test performance. KNN (Figure 8) exhibited steep early overfitting and limited validation improvement, consistent with its lower accuracy and unstable generalization.

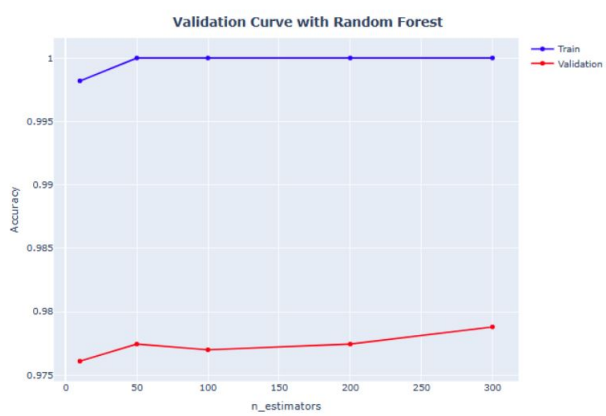

Figure 6: Validation Curve for Random Forest Predicting Democratic-Leaning Counties

Feature importance scores from the Random Forest model (Figure 6) clarify which variables most strongly influenced model predictions. The most dominant feature was `per_dem_20`, or the percentage of the county's vote that went to the Democratic candidate (Biden) in the 2020 presidential election. This variable alone accounted for over 60% of the total importance, indicating that prior voting behavior was, by far, the most predictive input. This result is expected, as political affiliation at the county level tends to be stable across election cycles. The 2020 vote share for Biden, especially with Kamala Harris on the ticket, likely reflects not only policy preferences but also strong racial and partisan alignment, particularly in areas with large Black populations and urban centers.

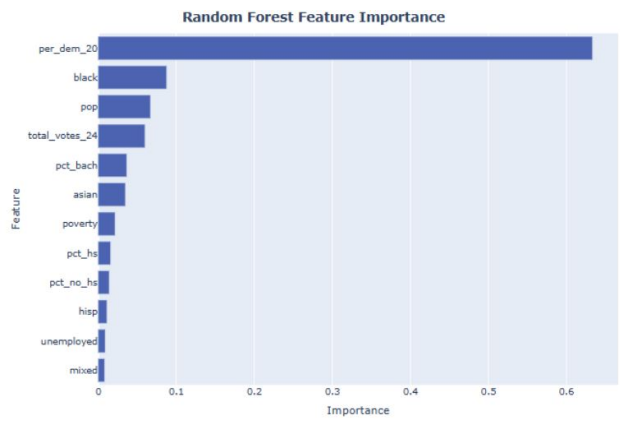

Figure 7: Feature Importance for Random Forest Partisan Lean Classifier

The second most important feature was pct_black, which aligns with long-standing patterns of racial polarization in U.S. electoral behavior.[9] Black voters have consistently supported Democratic candidates at high rates in federal elections, and counties with higher Black populations have historically leaned left, especially in the South and in urban districts. Furthermore, Kamala Harris's presence as the first Black woman presidential nominee likely strengthened the association between the share of the Black population and Democratic support, making it even more intuitive that racial composition would serve as a strong predictor in this context. The model's reliance on this feature suggests that racial composition is a structural component of partisan identity in the data.

Population size also ranked highly. This is likely due to the well-documented urban-rural divide in American politics, where denser, urbanized counties tend to vote Democratic, while rural counties skew Republican.[10] In practice, population may serve as a proxy for urbanicity, with large counties capturing major metropolitan regions that tend to lean Democratic due to demographic diversity, labor market structure, and political geography.

Other features, such as education levels and poverty rates, were included in the model but contributed relatively little to its predictions. While these factors are often correlated with political behav-

ior in individual-level analyses, they were less valuable in this aggregate classification task. This is likely because their effects are already captured indirectly through variables such as racial composition, urbanization, and, especially, prior vote share. Once a model has access to a county's 2020 partisan breakdown, secondary structural indicators add minimal marginal value.

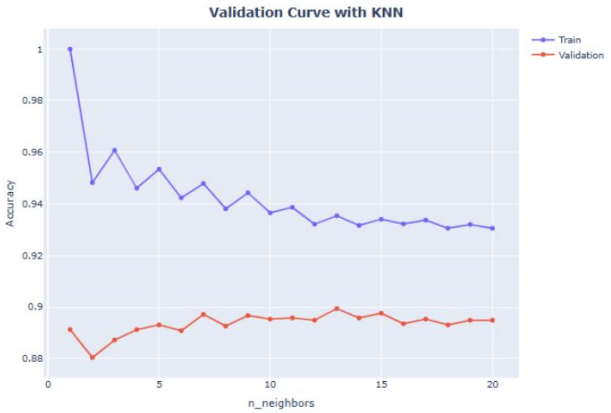

Figure 8: Validation Curve for KNN Predicting Democratic-Leaning Counties

In all, these results indicate that the classification task is largely tractable due to the stability of county-level voting behavior over time and the strength of a few high-signal features. Models that can leverage feature interactions (like Random Forest) or handle high-dimensional scaling (like Logistic Regression) were most successful. In contrast, KNN lacked the flexibility to accommodate these dynamics and suffered from high variance. The fact that SVM over-predicted the non-Democratic class further underscores how decision boundaries can become miscalibrated in the presence of strong class priors and unbalanced feature influence.

4.2.2 High Turnout (`total_votes_24`)

Four models—Logistic Regression, Random Forest, SVM, and KNN—were evaluated for their ability to classify counties as having high or low turnout based on demographic and socioeconomic features selected through forward stepwise selection.

Table 8: Test Set Classification Metrics for Predicting High vs. Low Voter Turnout

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.935	0.956	0.900	0.927
Random Forest	0.971	0.978	0.958	0.968
SVM	0.866	0.914	0.782	0.843
KNN	0.708	0.662	0.745	0.701

As shown in Table 5, Random Forest again achieved the highest test accuracy (0.971), followed by Logistic Regression (0.935). SVM performed moderately well (0.866), but struggled with recall, while KNN had the weakest overall performance (0.708), failing to generalize despite achieving reasonable recall.

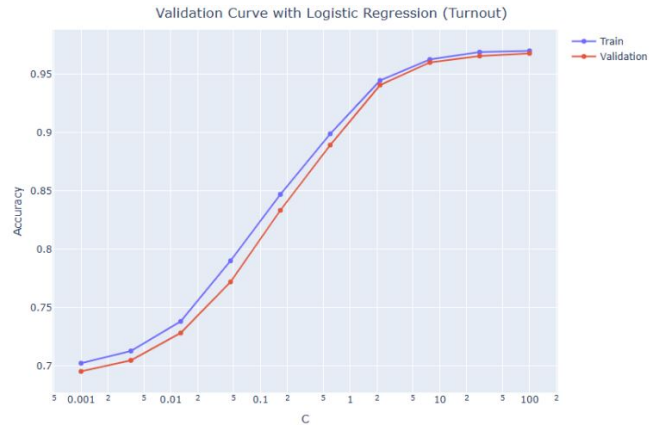

Figure 9: Validation Curve for Logistic Regression Predicting Turnout

Once again, Random Forest delivered the strongest results, with balanced precision and recall and an F1 score of 0.968. Logistic Regression followed closely behind, maintaining solid performance without overfitting. SVM (Figure 10) displayed high precision but lower recall, suggesting it was more conservative in assigning the high-turnout label. KNN's recall was higher than expected, given its accuracy; however, overall instability and a significant gap between training and validation performance (Figure 13) limited its reliability.

Figure 10: Validation Curve for SVM Predicting Turnout

Logistic Regression and SVM showed consistent improvement in validation accuracy as the regularization parameter C increased, eventually plateauing above 0.95 (Figures 9 and 10). This indicates that both models were able to fit the decision boundary more effectively as complexity increased. However, SVM's wider gap between training and validation accuracy suggests it was more prone to overfitting compared to Logistic Regression. Random Forest again achieved perfect training accuracy quickly (Figure 11), with slightly flatter and lower validation accuracy, indicating mild overfitting. KNN's sharp drop in validation accuracy across increasing k (Figure 12) was consistent with its test performance and confirmed that the model failed to generalize across the feature space.

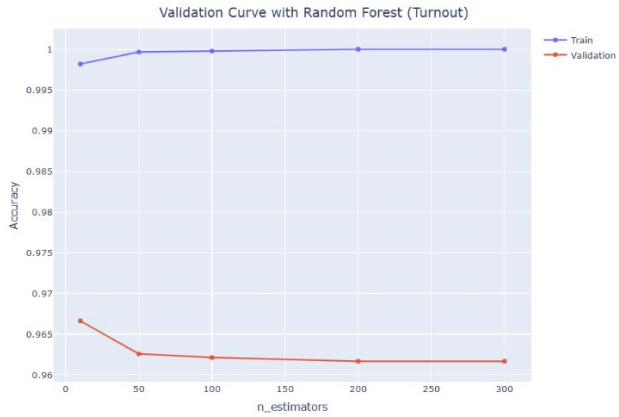

Figure 11: Validation Curve for Random Forest Predicting Turnout

Random Forest feature importance scores (Figure 12) indicate that population size (`pop`) was by far the most influential factor in predicting voter turnout. This aligns with expectations, as counties with larger populations often have better voter mobilization infrastructure, more competitive elections, and higher civic engagement.[11] The next most important feature was `pct_bach`, the share of residents with a bachelor's degree—an established predictor of voting behavior, especially in midterm and presidential elections. Higher educational attainment is consistently associated with a greater likelihood of voting, and its importance in this context reinforces this association.

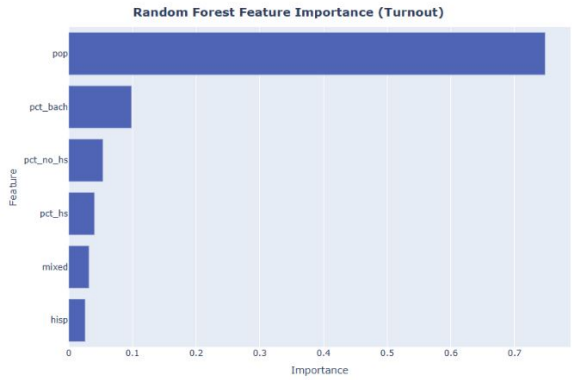

Figure 12: Feature Importance for Random Forest Turnout Classifier

Other selected features, including `pct_no_hs`, `pct_hs`, `hisp`, and `mixed`, had much lower importance, suggesting that, while race and education are both frequently associated with turnout in individual-level studies, the aggregate structure of the dataset places most predictive weight on population size and postsecondary educational attainment. Importantly, unlike the partisan classification task, there was no equivalent to `per_dem_20`—a near-deterministic predictor. As a result, model performance was slightly more variable, and feature contributions were more evenly distributed, with the exception of the population.

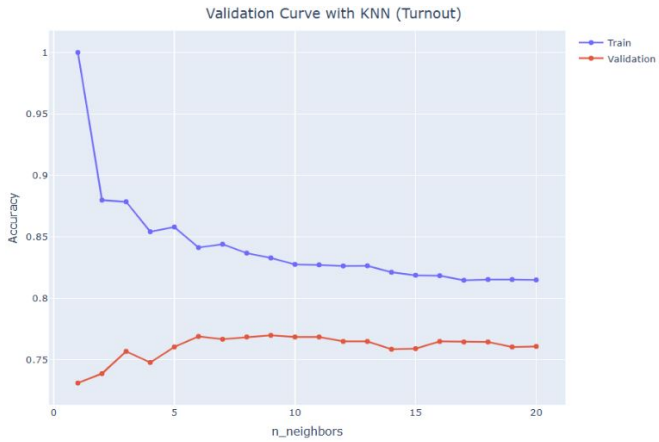

Figure 13: Validation Curve for KNN Predicting Turnout

Overall, predicting turnout proved slightly more challenging than classifying partisan lean, due to the absence of a single dominant predictor and the more diffuse nature of turnout behavior. However, ensemble and linear models still performed well, while instance-based methods, such as KNN, again suffered from high variance and overfitting.

5 Discussion

While model performance was strong across both regression and classification tasks, particularly in predicting Democratic vote share and overall turnout, these results should be interpreted within the bounds of a simulated environment. The strongest models leveraged stable historical patterns and high-signal features, such as prior vote share and population size. However, the structure of the task and the assumptions embedded in the data limit how directly these results can be generalized to live election-night contexts. Building on these results, future efforts could improve model realism and adaptability by incorporating real-time learning, time-aware vote share forecasting, and uncertainty quantification. These enhancements would better reflect the complexities of election night prediction, where incomplete data, reporting delays, and regional variation complicate decision-making.

5.1 Limitations

The 2024 election data were obtained from scraped results, including those from Fox News for 2024 and Fox News, Politico, and The New York Times for 2020. These are not official datasets and may include missing or inconsistent county-level data. This affects data quality and limits reliability across states. Further, the classification tasks employed median splits to define binary targets, which simplified the problem but also removed detail and may have obscured patterns in more competitive or marginal counties.

States do not report all their results simultaneously and often continue counting ballots well beyond election day. The split of states into test and training sets is thus entirely arbitrary and not reflective of the actual pattern of how results were reported in 2024. True election night models must account for counties that have partially reported results in their predictions.

Additionally, the use of Census demographic data to model voter demographics creates bias. The education and unemployment variables are reflective of a county's adult population, and the race, urbanicity, and median household income variables are reflective of a county's entire population. The voting population differs systematically from both of these populations; therefore, care must be taken when interpreting variables. It would be incorrect to say that counties with a higher proportion of Asian voters had higher voter turnout, given that the data used to construct the model measures the entire population of a county. It is possible that the model would be more accurate if the demographic data were reflective of the voting population, as obtained from a source such as a verified voter file matched with demographic data.

5.2 Future Works

One avenue for future work is to implement real-time learning, where predictions are updated continuously as new county-level results become available. Instead of treating the test set as static, the model would treat counties with early or partial returns as an evolving validation stream, enabling dynamic changes in predictions throughout the night. This better mirrors how live decision desks refine estimates as data trickles in. Another extension involves modeling the temporal trajectory of vote share using time series methods. For example, if Harris' vote share in a state is 60% at 9 PM and 58% at 10 PM, a model trained on historical vote reporting patterns could forecast a likely drop to 56% by 11 PM. This approach would help anticipate hour-by-hour changes, particularly in close races where reporting delays and ballot type (e.g., Democrat-heavy early votes vs. Republican-heavy same day votes) affect outcome trends. Simulating partial reporting is also essential, and a more robust testing framework could randomly mask portions of county-level data to emulate the staggered reporting of real election returns. This would enable the model to be evaluated under conditions that more accurately reflect what networks and campaigns encounter on election night, when decisions must be made with incomplete data. Finally, quantification of uncertainty could be critical– rather than reporting only point predictions, future versions should incorporate probabilistic frameworks to provide more comprehensive insights. Monte Carlo simulations, Bayesian modeling, or bootstrapping can be used to generate confidence intervals and posterior distributions over predicted outcomes, which would improve interpretability and help election analysts better communicate the inherent ambiguity that accompanies live forecasting.

6 Conclusion

Political organizations and the media alike have an incentive in predicting the winner of an election based on partial returns as early and accurately as possible. As a result, it is pertinent to study which model best predicts each county's turnout and partisan lean based on partial results that are reported earlier in the night. In other words, can data from states that report returns early in the night (e.g., Kentucky, Indiana, and Florida) be used to predict results in swing

states that report late in the night? How accurate are these predictions?

Linear regression models performed best in this scenario. The best linear regression models (LASSO for partisan lean and cross-validated linear regression for turnout) had lower errors, on average, than a more complex artificial neural network. The end result was a combined turnout and partisan lean model that correctly predicted the winner of each of the states in the test set.

Biden's 2020 vote share in a county was the strongest predictor of Harris's 2024 vote share in a county, while unemployment and population growth were negative predictors of her vote share. 2020 turnout and population growth were strongly, positively correlated with 2024 turnout, while the total population was negatively correlated with turnout, suggesting urban areas had lower voter turnout.

In classification tasks, an artificial neural network had the highest accuracy (98.1%) compared to other tested methods like k-nearest neighbors, logistic regression, or random forest. This suggests a more complex relationship between the predictor variables and whether or not a county had a higher-than-median Democratic vote share or turnout.

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